

Farhan Akhtar Makandar

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience in designing and deploying end-to-end AI systems spanning machine learning, deep learning, and large language models. Demonstrated ability to build production-grade AI pipelines including data ingestion, model training, evaluation, and deployment. Skilled in transformer architectures, retrieval-augmented generation, and intelligent system design. Research experience at NIT Calicut on neural network optimization for real-world engineering problems.

TECHNICAL SKILLS

- Programming: Python, SQL, C, R, HTML/CSS
- AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Transformers
- LLM & NLP: RAG Pipelines, Prompt Engineering, LLM Integration, Embeddings, Fine-Tuning
- Deep Learning: CNN, RNN, LSTM, GANs, Attention Mechanisms, SAINT
- Data Science: Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Feature Engineering
- Agentic AI: Vector Databases, Agentic Workflows, LangChain, RAG Pipelines
- Cloud & DevOps: AWS, Oracle Cloud, Git/GitHub, Cloudflare, Vercel

PROFESSIONAL EXPERIENCE

Research Intern - AI/ML - NIT Calicut

Jun 2024 - Dec 2024

- Developed and evaluated transformer-based models for complex predictive tasks using PyTorch, achieving rigorous performance benchmarks
- Collaborated on cutting-edge AI research focused on neural network optimization and deep learning applications for engineering problems
- Conducted comprehensive data analysis, feature engineering, and model validation using scientific computing libraries
- Assisted in documenting research findings and preparing technical reports for publication

PROJECTS

Agentic AI System - RAG Pipeline

Engineered end-to-end RAG pipeline: document retrieval using vector databases, LLM-based reasoning and response generation with context-aware prompt engineering. Integrated vector database for efficient semantic search.

Techn

Seismic Response Prediction - TFT

Developed Temporal Fusion Transformer achieving 22% improvement over traditional ML baselines (XGBoost, KNN) for seismic structural response prediction. Comprehensive EDA and model evaluation.

Techn

SAINT Model - Attention-Based Classification

Built Self-Attention and Intersample Attention model for tabular data classification with performance optimization through attention architecture.

Techn

EDUCATION

B.E. in Artificial Intelligence & Machine Learning

Yenepoya Institute of Technology, Moodbidri (VTU) | Expected 2026

CERTIFICATIONS

- IBM Python for Data Science (Credly Verified)
- Machine Learning with Python - IBM Cognitive Class
- Deep Learning Fundamentals - IBM Cognitive Class
- Accelerating Deep Learning with GPUs - IBM Cognitive Class
- Supervised Machine Learning - Coursera
- Building RAG Agents with LLMs - NVIDIA
- AWS Generative AI Essentials - AWS
- AWS Prompt Engineering - AWS
- Oracle Cloud Infrastructure 2025 AI Foundations Associate
- Accenture Data Analytics Simulation - Forage
- HTML Training - IIT Bombay